Companion XV: The Early-Universe Gravitational-Wave Spectrum in the Relaxation-Driven Cyclic Cosmology (RDCC)

Relaxation-Induced Tensor Damping, Sectoral Asymmetry, and CPT-Sector GW Signatures

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Abstract

The early-Universe gravitational-wave (GW) spectrum provides a uniquely sensitive probe of the microphysical processes that shaped the primordial plasma. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduced in Companion X modifies the evolution of tensor perturbations during the earliest stages of cosmic history. The relaxation rate $\Gamma(a) = \alpha(a)H(a)$ introduces an additional damping term in the tensor-mode equation of motion, suppressing the amplitude of primordial gravitational waves on scales that re-entered the horizon during the relaxation-dominated epoch.

Combined with the sectoral temperature asymmetry between the two CPT-conjugate sectors, this leads to a characteristic sectoral GW asymmetry and a relaxation-imprinted suppression feature in the stochastic gravitational-wave background (SGWB). The colder CPT sector experiences weaker damping and produces a distinct tensor spectrum, while the visible sector exhibits a relaxation-induced suppression at high frequencies.

In this Companion we derive the relaxation-corrected tensor-mode evolution, compute the resulting GW transfer function, and obtain the full early-Universe GW spectrum in RDCC. We incorporate the modernized sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II. The results establish the SGWB as a central observational probe of the relaxation-driven dynamics that define the RDCC framework.

1 Motivation

The stochastic gravitational-wave background (SGWB) generated in the early Universe provides a uniquely sensitive probe of high-energy physics. In standard cosmology, the SGWB is determined by the primordial tensor spectrum and its subsequent evolution through radiation and matter domination. The resulting spectrum is smooth, nearly scale-invariant, and free of dynamical features except for small steps associated with changes in the relativistic degrees of freedom.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The universal relaxation mechanism introduced in Companion X modifies the evolution of super-horizon tensor modes during the earliest stages of cosmic history. When the relaxation rate satisfies

$$\Gamma(a) = \alpha(a)H(a) \gtrsim H(a), \tag{1}$$

tensor modes experience an additional damping term that suppresses their amplitude before horizon re-entry. This effect leaves a characteristic imprint on the SGWB: a smooth, power-law suppression at frequencies corresponding to modes that re-entered during the relaxation-dominated epoch.

A second key ingredient is the sectoral temperature asymmetry between the two CPT-conjugate sectors. As shown in Companion X, the temperature ratio

$$\frac{T_B}{T_A} \simeq 0.5, \tag{2}$$

implies different relaxation histories, horizon masses, and damping scales in the two sectors. Tensor modes in the colder CPT sector experience weaker damping, leading to a sectoral GW asymmetry.

This Companion develops the early-Universe gravitational-wave sector of RDCC, incorporating the modernized sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II. The SGWB emerges as a central observational probe of the relaxation-driven dynamics that define the RDCC framework.

The structure of this Companion is as follows. Section 2 reviews tensor modes in standard cosmology. Section 3 derives the relaxation-induced tensor damping in RDCC. Section 4 constructs the relaxation-corrected transfer function. Section 5 presents the full early-Universe GW spectrum. Section 6 develops the observational predictions, and Section 7 provides the conclusion.

2 Tensor Modes in Standard Cosmology

In standard cosmology, primordial tensor perturbations obey the equation

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \tag{3}$$

with two distinct dynamical regimes:

- **Super-horizon regime** ($k < aH$): Tensor modes freeze out with constant amplitude, $\dot{h}_k \approx 0$.
- **Sub-horizon regime** ($k > aH$): Tensor modes oscillate with physical frequency k/a and decay as a^{-1} due to Hubble friction.

The primordial tensor spectrum is commonly parameterized as

$$\mathcal{P}_h(k) = A_t \left(\frac{k}{k_*}\right)^{n_t}, \tag{4}$$

where A_t is the tensor amplitude, n_t is the tensor tilt, and k_* is a pivot scale.

The present-day gravitational-wave energy density per logarithmic interval in wavenumber is

$$\Omega_{\rm GW}(k) = \frac{1}{12}\left(\frac{k}{a_0H_0}\right)^2 \mathcal{P}_h(k)\, T^2(k), \tag{5}$$

where $T(k)$ is the tensor transfer function.

In standard cosmology, the transfer function is determined by:

- horizon re-entry during radiation or matter domination,
- changes in the relativistic degrees of freedom,

- the transition from radiation to matter domination.

Crucially, **standard cosmology contains no additional damping beyond Hubble friction**. The SGWB is therefore smooth and featureless, making it exceptionally sensitive to any new early-Universe physics.

This provides the baseline against which the relaxation-induced tensor damping of RDCC must be compared, developed in Section 3.

3 Relaxation-Induced Tensor Damping

In the Relaxation-Driven Cyclic Cosmology (RDCC), the evolution of tensor perturbations is modified by the universal relaxation mechanism introduced in Companion X. The relaxation rate

$$\Gamma(a) = \alpha(a)H(a) \tag{6}$$

acts as an additional friction term in the tensor-mode equation of motion. This modifies the dynamics of super-horizon tensor modes during the relaxation-dominated epoch and imprints a characteristic suppression feature on the stochastic gravitational-wave background (SGWB).

The tensor-mode equation in RDCC becomes

$$\ddot{h}_k + (3H + \Gamma)\,\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \tag{7}$$

which can be written as

$$\ddot{h}_k + (3 + \alpha)H\dot{h}_k + \frac{k^2}{a^2}h_k = 0. \tag{8}$$

3.1 Super-Horizon Evolution

For modes outside the horizon ($k < aH$), the gradient term is negligible and the equation reduces to

$$\ddot{h}_k + (3 + \alpha)H\dot{h}_k \approx 0. \tag{9}$$

In standard cosmology, tensor modes freeze out with constant amplitude. In RDCC, the additional relaxation term suppresses the amplitude according to

$$h_k(a) \propto a^{-\alpha/2}, \tag{10}$$

implying that modes which remain super-horizon during the relaxation-dominated epoch experience a smooth, power-law suppression.

This suppression is the tensor analogue of the relaxation-induced damping of scalar perturbations derived in Companion X.

3.2 Sub-Horizon Evolution

Once a mode re-enters the horizon ($k > aH$), the oscillatory term dominates and the solution takes the form

$$h_k(a) \propto \frac{1}{a}\exp\left[-\frac{1}{2}\int \alpha(a)\,d\ln a\right]\cos\left(k\int \frac{da}{a^2H}\right). \tag{11}$$

The usual a^{-1} decay from Hubble friction is now multiplied by an additional exponential suppression factor sourced by the relaxation dynamics. This factor depends on the integrated relaxation history of the mode.

3.3 Relaxation-Dominated Epoch

The relaxation-dominated regime is defined by

$$\Gamma(a) \gtrsim H(a), \qquad \text{or equivalently} \qquad \alpha(a) \gtrsim 1. \tag{12}$$

Modes that re-enter the horizon during this epoch experience the strongest suppression. The corresponding wavenumbers satisfy

$$k_{\rm rel} \sim a_{\rm rel} H_{\rm rel}, \tag{13}$$

where $a_{\rm rel}$ denotes the scale factor at which $\alpha(a)$ transitions below unity.

Modes with $k > k_{\rm rel}$ re-enter during the relaxation-dominated epoch and are therefore suppressed, while modes with $k < k_{\rm rel}$ re-enter later and retain their standard amplitude.

This produces a smooth, power-law suppression feature in the SGWB.

3.4 Sectoral Tensor Asymmetry

The sectoral temperature asymmetry derived in Companion X,

$$\frac{T_B}{T_A} \simeq 0.5, \tag{14}$$

implies different relaxation histories in the two sectors:

$$\alpha_B(a) < \alpha_A(a). \tag{15}$$

As a result:

- tensor modes in the visible sector experience stronger damping,
- tensor modes in the CPT sector experience weaker damping,
- the SGWB exhibits a sectoral asymmetry in amplitude and spectral shape.

This sectoral tensor asymmetry is a distinctive and testable prediction of RDCC and forms the basis for the relaxation-corrected transfer function developed in Section 4.

4 Relaxation-Corrected Transfer Function

The transfer function encodes the evolution of tensor modes from horizon re-entry to the present day. In standard cosmology, the transfer function is determined by the expansion history and the changes in the relativistic degrees of freedom. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces an additional suppression factor that modifies the transfer function at high frequencies.

The relaxation-corrected transfer function can be written as

$$T^2(k) = T^2_{\rm std}(k)\, T^2_{\rm rel}(k), \tag{16}$$

where $T_{\rm std}(k)$ is the standard transfer function and $T_{\rm rel}(k)$ captures the relaxation-induced suppression.

4.1 Standard Transfer Function

The standard transfer function is given by

$$T_{\rm std}^2(k) = \Omega_{r,0}\left[1 + \frac{4}{3}\left(\frac{k}{k_{\rm eq}}\right) + \frac{5}{2}\left(\frac{k}{k_{\rm eq}}\right)^2\right]^{-1}, \tag{17}$$

where $k_{\rm eq}$ is the wavenumber corresponding to matter–radiation equality and $\Omega_{r,0}$ is the present-day radiation density fraction.

This expression captures the standard suppression of tensor modes that re-enter during matter domination.

4.2 Relaxation-Induced Suppression Factor

The relaxation-induced suppression factor arises from the additional damping term in the tensor-mode equation of motion. For a mode with wavenumber k, the suppression is given by

$$T_{\rm rel}(k) = \exp\left[-\frac{1}{2}\int_{\ln a_{\rm hc}}^{\ln a_{\rm end}} \alpha(a)\, d\ln a\right], \tag{18}$$

where $a_{\rm hc}$ is the scale factor at horizon crossing and $a_{\rm end}$ marks the end of the relaxation-dominated epoch.

Modes that re-enter the horizon during the relaxation-dominated epoch experience the largest suppression. Modes that re-enter later are unaffected.

4.3 Characteristic Relaxation Scale

The characteristic relaxation scale is defined by the wavenumber

$$k_{\rm rel} = a_{\rm rel} H_{\rm rel}, \tag{19}$$

where $a_{\rm rel}$ is the scale factor at which $\alpha(a)$ transitions below unity. Modes with $k > k_{\rm rel}$ re-enter during the relaxation-dominated epoch and are therefore suppressed, while modes with $k < k_{\rm rel}$ retain their standard amplitude.

The relaxation-induced suppression factor can be approximated as

$$T_{\rm rel}(k) \simeq \left(\frac{k}{k_{\rm rel}}\right)^{-\alpha_{\rm eff}}, \qquad k > k_{\rm rel}, \tag{20}$$

where $\alpha_{\rm eff}$ is an effective relaxation coefficient that captures the integrated relaxation history.

4.4 Sectoral Transfer Functions

The sectoral temperature asymmetry derived in Companion X,

$$\frac{T_B}{T_A} \simeq 0.5, \tag{21}$$

implies different relaxation histories in the two sectors. The relaxation coefficients satisfy

$$\alpha_B(a) < \alpha_A(a), \tag{22}$$

leading to sector-dependent transfer functions:

$$T_{\rm rel}^{(B)}(k) > T_{\rm rel}^{(A)}(k). \tag{23}$$

The visible sector therefore exhibits a stronger high-frequency suppression, while the CPT sector retains a larger tensor amplitude.

This sectoral asymmetry is a distinctive and testable prediction of RDCC and forms the basis for the full early-Universe GW spectrum developed in Section 5.

5 Full Early-Universe GW Spectrum

The relaxation-corrected transfer function derived in Section 4 allows us to construct the full early-Universe gravitational-wave (GW) spectrum in the Relaxation-Driven Cyclic Cosmology (RDCC). The present-day GW energy density per logarithmic interval in wavenumber is given by

$$\Omega_{\rm GW}(k) = \frac{1}{12}\left(\frac{k}{a_0 H_0}\right)^2 \mathcal{P}_h(k)\, T^2_{\rm std}(k)\, T^2_{\rm rel}(k), \tag{24}$$

where the relaxation-induced suppression factor $T_{\rm rel}(k)$ encodes the integrated relaxation history of each mode.

The resulting spectrum exhibits three characteristic regimes.

5.1 Low-Frequency Regime

Modes with $k < k_{\rm rel}$ re-enter the horizon after the relaxation-dominated epoch. Their amplitude is unaffected by the relaxation mechanism, and the spectrum reduces to the standard result:

$$\Omega_{\rm GW}(k) \simeq \Omega^{\rm std}_{\rm GW}(k), \qquad k < k_{\rm rel}. \tag{25}$$

This regime is identical in both sectors and provides a baseline for comparison.

5.2 Intermediate Regime

Modes with $k \sim k_{\rm rel}$ re-enter the horizon near the end of the relaxation-dominated epoch. These modes experience partial suppression, leading to a smooth transition between the standard and relaxation-dominated regimes. The spectrum takes the form

$$\Omega_{\rm GW}(k) \simeq \Omega^{\rm std}_{\rm GW}(k)\left(\frac{k}{k_{\rm rel}}\right)^{-2\alpha_{\rm eff}}, \qquad k \sim k_{\rm rel}. \tag{26}$$

This regime encodes the detailed relaxation history and is sensitive to the shape of $\alpha(a)$.

5.3 High-Frequency Regime

Modes with $k > k_{\rm rel}$ re-enter during the relaxation-dominated epoch and experience the strongest suppression. The spectrum becomes

$$\Omega_{\rm GW}(k) \simeq \Omega^{\rm std}_{\rm GW}(k)\left(\frac{k}{k_{\rm rel}}\right)^{-2\alpha_{\rm eff}}, \qquad k > k_{\rm rel}. \tag{27}$$

This power-law suppression is the defining signature of relaxation-induced tensor damping.

5.4 Sectoral GW Spectra

The sectoral temperature asymmetry derived in Companion X implies different relaxation histories in the two sectors:

$$\alpha_B(a) < \alpha_A(a). \tag{28}$$

As a result:

- the visible sector exhibits a stronger high-frequency suppression,
- the CPT sector retains a larger tensor amplitude,
- the SGWB displays a sectoral asymmetry in amplitude and spectral slope.

The full sectoral spectra are therefore

$$\Omega_{\rm GW}^{(A)}(k) = \Omega_{\rm GW}^{\rm std}(k) \left(\frac{k}{k_{\rm rel}^{(A)}}\right)^{-2\alpha_{\rm eff}^{(A)}}, \tag{29}$$

$$\Omega_{\rm GW}^{(B)}(k) = \Omega_{\rm GW}^{\rm std}(k) \left(\frac{k}{k_{\rm rel}^{(B)}}\right)^{-2\alpha_{\rm eff}^{(B)}}, \tag{30}$$

with

$$\alpha_{\rm eff}^{(B)} < \alpha_{\rm eff}^{(A)}, \qquad k_{\rm rel}^{(B)} < k_{\rm rel}^{(A)}. \tag{31}$$

These expressions define the complete early-Universe GW spectrum in RDCC and form the basis for the observational predictions developed in Section 6.

6 Observational Predictions

The relaxation-induced tensor damping and the sectoral temperature asymmetry of the Relaxation-Driven Cyclic Cosmology (RDCC) lead to a number of distinctive observational signatures in the stochastic gravitational-wave background (SGWB). These signatures provide a direct probe of the relaxation dynamics, the bipartite CPT structure, and the early-Universe thermodynamics developed in Companions X, XIII, and XIV.

6.1 High-Frequency Suppression

The defining prediction of RDCC is a smooth, power-law suppression of the SGWB at high frequencies:

$$\Omega_{\rm GW}(k) \propto \left(\frac{k}{k_{\rm rel}}\right)^{-2\alpha_{\rm eff}}, \qquad k > k_{\rm rel}. \tag{32}$$

This suppression arises from the relaxation-induced damping of tensor modes that re-enter the horizon during the relaxation-dominated epoch. The scale $k_{\rm rel}$ marks the transition between the standard and relaxation-modified regimes.

6.2 Sectoral GW Asymmetry

The sectoral temperature ratio

$$\frac{T_B}{T_A} \simeq 0.5 \tag{33}$$

implies different relaxation histories in the two sectors. As a result:

- the visible sector exhibits a stronger high-frequency suppression,
- the CPT sector retains a larger tensor amplitude,
- the SGWB displays a sector-dependent spectral slope.

This sectoral asymmetry is a distinctive and testable prediction of RDCC.

6.3 Connection to PBH Physics

The same relaxation dynamics that suppress tensor modes also suppress PBH formation in the visible sector and enhance PBH formation in the CPT sector. The SGWB and the PBH sector therefore provide complementary probes of the relaxation-driven early-Universe physics.

The characteristic scales $k_{\rm rel}^{(A/B)}$ are directly related to the PBH mass scales derived in Companion XIV.

6.4 Experimental Prospects

Future gravitational-wave observatories will be able to probe the relaxation-induced features of the SGWB:

- **LISA**: sensitive to the intermediate regime near k_{rel},
- **DECIGO/BBO**: sensitive to the high-frequency suppression,
- **Einstein Telescope / Cosmic Explorer**: complementary constraints at lower frequencies.

A detection of a smooth, power-law suppression at high frequencies would provide strong evidence for relaxation-driven tensor damping.

6.5 Consistency with RDCC Thermodynamics

The SGWB predictions are fully consistent with the sectoral thermodynamics developed in Companion XIII and the PBH sector developed in Companion XIV. The relaxation-induced tensor suppression, the sectoral asymmetry, and the PBH mass hierarchy all arise from the same underlying relaxation dynamics and the bipartite CPT structure of RDCC.

7 Conclusion

In this Companion we have developed the early-Universe gravitational-wave (GW) sector of the Relaxation-Driven Cyclic Cosmology (RDCC). The universal relaxation mechanism introduced in Companion X modifies the evolution of tensor perturbations during the earliest stages of cosmic history, introducing an additional damping term that suppresses the amplitude of primordial gravitational waves on scales that re-entered the horizon during the relaxation-dominated epoch.

The resulting stochastic gravitational-wave background (SGWB) exhibits a smooth, power-law suppression at high frequencies, with the characteristic scale set by the relaxation transition k_{rel}. This suppression is the tensor analogue of the relaxation-induced damping of scalar perturbations and provides a direct probe of the relaxation dynamics that define the RDCC framework.

The sectoral temperature asymmetry between the two CPT-conjugate sectors leads to different relaxation histories and therefore to a sectoral GW asymmetry. Tensor modes in the visible sector experience stronger damping, while modes in the CPT sector retain a larger amplitude. This sectoral asymmetry is a distinctive and testable prediction of RDCC.

The full early-Universe GW spectrum constructed in this Companion incorporates the modernized sectoral thermodynamics from Companion XIII, the CPT-sector PBH interpretation from Companion XIV, and the tensor-modulation structure from Companion II. Together, these results establish the SGWB as a central observational probe of the relaxation-driven early-Universe physics that underlies the RDCC framework.

Future gravitational-wave observatories such as LISA, DECIGO, BBO, and the Einstein Telescope will be able to test the relaxation-induced suppression feature and the sectoral GW asymmetry. A detection of these signatures would provide strong evidence for the relaxation mechanism and the bipartite CPT structure that define RDCC.